

Grand canonical simulation of a Lennard-Jones system

1.
 - a. Unzip and untar archive.tar.gz
(gunzip archive.tar.gz, tar -xvf archive.tar)
 - b. Compile the simulation program (make mc)
 - c. Copy the executable „LJ“ to the SIMULATION folder and start a simulation at $T=0.97$ und $\mu=-2.865$ (./LJ eingabe &). The simulation should take at most 30 minutes. In the meantime work on exercise 2.
2. Go to AUSWERTUNG
 - a. Make a histogram (x-axis: number of particles N (second column), y-axis: number of occurrences) and plot the result. On what kind of simulation is the data based on?
 - b. Generate the true (unweighted) probability distribution $P(N)$ and the free energy $F(N)$ and plot them
(Hint: $w(N)$ is located in the fifteenth column.)
 - c. Determine the average energy per particle (fourth column in histo.dat) in the gas and the liquid phase. Explain the qualitative difference between the two.
3. Go to SIMULATION
 - a. Determine the chemical potential at coexistence and the coexistence densities.
 - b. Sketch the phase diagram $T(\rho)$ for the Lennard-Jones system based on results from 2 and 3.
 - c. Determine a weight function $w(N)$ for a simulation at $T=0.95$. At which chemical potential would you carry out the simulation?
 - d. You would like to determine the coexistence densities of a LJ system at $T=0.75$. How would you do that?